

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****Summer Examination – 2023****Course: T.Y.B. Tech.      Branch: Mechanical Engineering      Semester: VI****Subject Code & Name: BTMOE605C Energy Conservation and Management****Max Marks: 60****Date: 21-07-2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                      | (Level/CO)                   | Marks     |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------|------------|-----------------|----------------|-----------|---|---------|---|---------|---|---------|---|---------|---|---------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                          |                              | <b>12</b> |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| A) What is the need of energy management? State principles of energy management.                                                                                                                                                                                                                     | Level 2<br>Understand<br>CO1 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| B) What is the difference between Energy conservation and energy efficiency? Explain with suitable examples.                                                                                                                                                                                         | Level 2<br>Understand<br>CO1 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| C) Describe the energy conservation opportunities in small and medium scale industries.                                                                                                                                                                                                              | Level 2<br>Understand<br>CO1 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                           |                              | <b>12</b> |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| A) Describe various types of energy audits.                                                                                                                                                                                                                                                          | Level 2<br>Understand<br>CO2 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| B) Consider a project, which has the following cash flow stream. The cost of capital, $k$ , for the firm is 10 %.                                                                                                                                                                                    | Level 3<br>Apply<br>CO3      | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| <table><tr><th>Investment</th><th>Rs. (10,00,000)</th></tr><tr><th>Saving in Year</th><th>Cash Flow</th></tr><tr><td>1</td><td>200,000</td></tr><tr><td>2</td><td>200,000</td></tr><tr><td>3</td><td>300,000</td></tr><tr><td>4</td><td>300,000</td></tr><tr><td>5</td><td>350,000</td></tr></table> |                              |           | Investment | Rs. (10,00,000) | Saving in Year | Cash Flow | 1 | 200,000 | 2 | 200,000 | 3 | 300,000 | 4 | 300,000 | 5 | 350,000 |
| Investment                                                                                                                                                                                                                                                                                           | Rs. (10,00,000)              |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| Saving in Year                                                                                                                                                                                                                                                                                       | Cash Flow                    |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| 1                                                                                                                                                                                                                                                                                                    | 200,000                      |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| 2                                                                                                                                                                                                                                                                                                    | 200,000                      |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| 3                                                                                                                                                                                                                                                                                                    | 300,000                      |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| 4                                                                                                                                                                                                                                                                                                    | 300,000                      |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| 5                                                                                                                                                                                                                                                                                                    | 350,000                      |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| Calculate the Net Present Value of this project.                                                                                                                                                                                                                                                     |                              |           |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| C) Enlist different instruments used for Energy auditing and explain any one of them.                                                                                                                                                                                                                | Level 2<br>Understand<br>CO2 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                          |                              | <b>12</b> |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| A) Explain energy conservation techniques in Boiler and Furnace.                                                                                                                                                                                                                                     | Level 2<br>Understand<br>CO2 | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |
| B) Explain simple payback period with advantages and disadvantages. Calculate the payback period for following example-                                                                                                                                                                              | Level 3<br>Apply<br>CO3      | 6         |            |                 |                |           |   |         |   |         |   |         |   |         |   |         |

A chemical plant is planning to install continuous deodorizer costing Rs. 60 Lakh for purchasing and installation. It is going to cost Rs. 1.5 Lakh per year for its operation and maintenance. This deodorizer is expected to save Rs. 20 Lakh per year through saving the steam consumption as compared to batch deodorizer.

- |                                                                           |                              |   |
|---------------------------------------------------------------------------|------------------------------|---|
| C) Explain energy conservation techniques in steam and condenser systems. | Level 2<br>Understand<br>CO2 | 6 |
|---------------------------------------------------------------------------|------------------------------|---|

**Q.4 Solve Any Two of the following.**

- |                                                                                                                                                                                                                                                                                |                              |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---|
| A) Explain Topping cycle and Bottoming cycle with block diagram.                                                                                                                                                                                                               | Level 2<br>Understand<br>CO4 | 6 |
| B) What is the purpose of insulators? Enlist name of different insulating material.                                                                                                                                                                                            | Level 2<br>Understand<br>CO4 | 6 |
| C) What is the refrigeration load in TR when 15 m <sup>3</sup> /hr of water is cooled from 21°C to 15°C?<br>If the compressor motor draws 29 kW, chilled water pump draws 4.6 kW, condenser water pump draws 6.1 kW and Cooling Tower fan draws 2.7 kW, what is overall kW/TR? | Level 3<br>Apply<br>CO4      | 6 |

**Q.5 Solve Any Two of the following.**

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---|
| A) What is load factor? Explain load factor with load curve analysis.                                                                                                                                                                                                                                                                                                                                                                           | Level 2<br>Understand<br>CO5 | 6 |
| B) Explain the concept of power factor in electrical system, with its mathematical formula. State the benefits of power factor improvement.                                                                                                                                                                                                                                                                                                     | Level 2<br>Understand<br>CO5 | 6 |
| C) A chemical industry has kW consumption of 812 kW with demand of 1160 kVA. To improve the power factor & avoid penalty, the company has added 410 kVAr of reactive power at motor end. Calculate<br>1. Previous Power factor<br>2. Previous Reactive power in the system in kVAr<br>3. Reduced Reactive power in the system in kVAr<br>4. Reduced kVA demand in the system, if kW consumption remains unchanged.<br>5. Improved Power Factor. | Level 3<br>Apply<br>CO5      | 6 |

\*\*\* End \*\*\*